Island Chill

Manufacturing Execution System

Functions & Processes

A comprehensive guide to system modules, workflows, and role-based functionality

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1. System Overview

The Island Chill Manufacturing Execution System (MES) is a browser-based production operations platform designed for beverage manufacturing facilities. It connects shop-floor activities with management oversight through a role-based interface that presents only the functions relevant to each user.

1.1 Key Capabilities

Work Order Management — Create, assign, execute, and track production jobs across multiple filling lines.

Quality Control — Perform parameter checks at defined intervals with pass/fail recording and trend tracking.

Incident Management — Report, investigate, and resolve production incidents with structured checklists.

Maintenance Scheduling — Plan and execute pre-start and periodic maintenance with clearance workflows.

BOM Requisitions — Manage raw material flow from store to production based on approved Bills of Materials.

Role-Based Access — Nine built-in roles with granular permission control across 22 submodules.

1.2 Technology Stack

Layer	Technology	Purpose
Frontend	React 19 + TypeScript + Vite	User interface and state management
Styling	Tailwind CSS + shadcn/ui	Responsive, theme-aware components
State	Zustand + localStorage	Client-side state and persistence
Routing	React Router v7	Navigation and route guards
Animation	Framer Motion	Page transitions and micro-interactions
Backend (planned)	tRPC + Drizzle ORM + MySQL	API layer and database (future phase)

2. System Architecture

The MES follows a layered architecture with clear separation between presentation, application logic, and data layers. The diagram below illustrates the major components and their relationships.

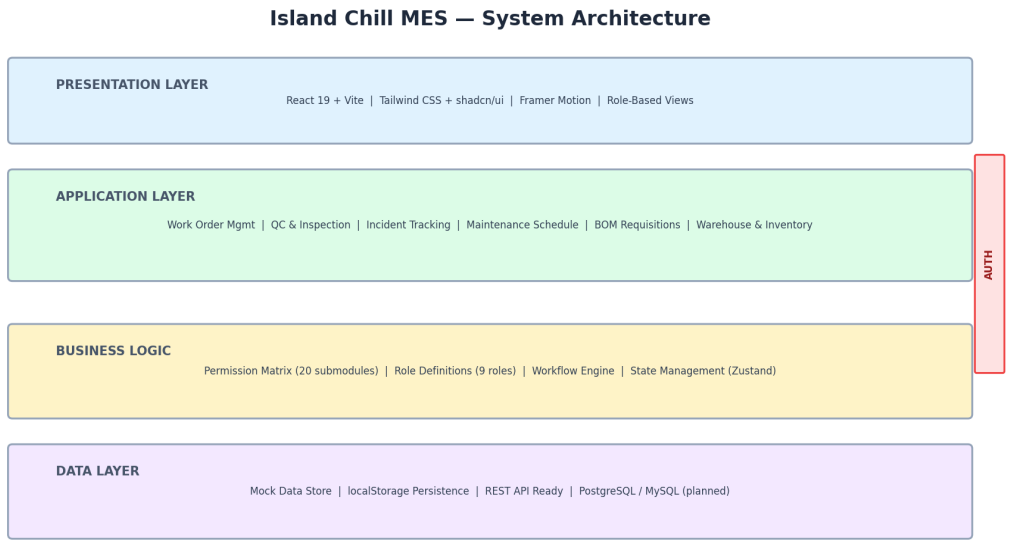


Figure 1: Island Chill MES System Architecture

2.1 Presentation Layer

The presentation layer is built with React 19 and TypeScript, compiled via Vite for fast development and optimized production builds. Tailwind CSS provides utility-first styling with a dark mode toggle. shadcn/ui components ensure consistent UI patterns. Framer Motion adds smooth page transitions and animated list updates.

2.2 Application Layer

The application layer contains all business modules: Work Order Management, Quality Control, Incident Tracking, Maintenance Scheduling, BOM Requisitions, and Warehouse operations. Each module is self-contained with its own pages, components, and data requirements.

2.3 Business Logic Layer

Zustand stores manage application state with localStorage persistence for theme preferences. The permission engine evaluates user roles against a 22-submodule matrix, controlling both menu visibility and route access. Route guards (RequirePermission) prevent unauthorized navigation.

2.4 Data Layer

Currently, the system uses a comprehensive mock data store for demonstration. In production, this layer will connect to a relational database (MySQL/PostgreSQL) via a tRPC API layer with Drizzle ORM. All data interfaces are already defined and ready for backend migration.

3. Module Descriptions

3.1 Work Order Management

The Work Order module is the production scheduling and execution hub. It manages the full lifecycle of a manufacturing job from creation through completion.

Function	Description	Roles
Create WO	Define product, variant, quantity, filling line, and schedule	Planner, Factory Manager
View WO List	Filterable table of all WOs with status badges	All roles
Execute WO	Track progress, update produced quantity, complete stages	Operator, Prod. Supervisor
Approve WO	Review and approve pending WOs before execution	Factory Manager
Cancel WO	Abort a WO with reason documentation	Factory Manager, Planner
WO Detail	Full timeline, status history, related QC checks and incidents	All roles

WO Status States: Pending — Created but not yet approved. Approved — Cleared for execution. In Progress — Production is active. Completed — All production and QC steps done. Cancelled — Aborted before or during execution.

3.2 Quality Control (QC)

The QC module ensures every production batch meets defined quality parameters. Checks are configured per product and scheduled at regular intervals during production.

Function	Description	Roles
Pending Checks	List of due checks with parameter name, range, and schedule	QC Inspector
Perform Check	Enter measured value, select Pass/Fail, add notes	QC Inspector
QC History	Completed checks with filter by pass/fail and date range	QC Inspector, Manager
QC Plans	View the check schedule for each product variant	QC Inspector, Manager
Forms Library	Standardized digital forms for consistent documentation	QC Inspector

3.3 Incident Management

The Incident module captures deviations from normal operations. It supports three permission levels (View, Report, Investigate) and provides a structured 10-step investigation checklist.

Function	Description	Roles
View Incidents	Browse incident list with severity, status, and assignment	All roles (View)
Report Incident	Raise new incidents with line, type, severity, and description	Operator, Supervisor
Investigate	Full investigation with checklist, root cause, photos, resolution	Maint. Sup., Prod. Sup., Factory Mgr.
Assign	Assign investigator and set priority	Production Supervisor, Factory Manager
Incident Detail	Full timeline, investigation notes, evidence photos	All (View only)

Root Cause Categories: Equipment Failure, Human Error, Material Defect, Process Deviation, Environmental Factor, Design Flaw, Training Gap, External Supplier, Other. Investigators must complete all 10 checklist items before resolving an incident.

3.4 Maintenance & Clearance

Maintenance is split into scheduled maintenance tasks and per-WO pre-start clearance. Both ensure equipment is safe and ready before production begins.

Function	Description	Roles
Scheduled Tasks	Recurring maintenance: Pre-start, Post-prod, Weekly, Monthly, Breakdown	Maintenance Supervisor
Task Checklist	Per-task itemized checks with Pass/Fail recording	Maintenance Supervisor
Pre-Start Clearance	WO-level maintenance approval before production starts	Maintenance Supervisor
Overdue Alerts	Visual indicators for tasks past their due date	All roles (View)
Maint. History	Completed tasks with timestamps and technician records	Maintenance Supervisor, Manager

3.5 Warehouse & Inventory

The Warehouse module tracks raw materials, packaging, and finished goods inventory. It provides low-stock alerts and links directly to BOM requisitions for material requests.

Function	Description	Roles
Inventory View	Stock levels by SKU with store location and expiry dates	Warehouse Supervisor
Low Stock Alerts	Automatic alerts when inventory falls below reorder point	Warehouse Supervisor, Manager
Goods Receipt	Record incoming materials with lot numbers and quantities	Warehouse Supervisor
Stock Movements	Track transfers, consumption, and adjustments	Warehouse Supervisor
Pick Requests	View and fulfill material pick lists from BOM requisitions	Warehouse Supervisor, Mixer

3.6 BOM Requisitions (Ingredient Mixer)

The Ingredient Mixer role manages the flow of raw materials from the warehouse store to the production floor. When a Planner approves a BOM, the system automatically generates a requisition slip that the Mixer uses to fetch, prepare, and deliver ingredients.

Status	Description	Action By
Draft	Requisition created, awaiting approval	System / Planner
Approved	BOM approved, ready for picking	Planner
Picking	Mixer has started fetching ingredients	Ingredient Mixer
Prepared	All ingredients fetched and verified	Ingredient Mixer
Delivered	Ingredients handed over to production	Ingredient Mixer

Each requisition includes a pick list with ingredient name, quantity, unit of measure, store location, lot number, and expiry date. The Mixer verifies each item before marking it prepared.

3.7 Administration

The Administration module is available to Administrators and Factory Managers. It provides system-wide configuration, user management, and reporting capabilities.

Module	Description	Status
Lines & Equipment	Register production lines and equipment with serial numbers and capacities	Planned
Products	Manage product catalog, variants, SKUs, and packaging configurations	Planned
BOMs	Create and manage Bills of Materials per product variant	Planned
Forms	QC and operational form template library	Planned
Workflow	Visual workflow designer and approval chain configuration	Planned
Users & Privileges	User accounts, role assignments, and permission matrix	Active

Reports	Production efficiency, quality trends, incident analytics	Planned
Settings	Company profile, shift schedules, integrations, data retention	Planned

4. Permission Matrix & Roles

The system defines 9 built-in roles with granular access across 22 permission submodules. Each role has a specific set of permissions that determines which menu items are visible, which routes are accessible, and which actions are allowed.

Role	Description	Key Permissions
Operator	Production floor worker	WO View, WO Execute, Incident Report, Dashboard
QC Inspector	Quality assurance staff	QC Inspect, QC View, WO View, Dashboard
Maintenance Sup.	Equipment maintenance lead	Maint. Schedule, Maint. View, Incident Investigate, Dashboard
Warehouse Sup.	Inventory and logistics	Warehouse, Stock View, Pick Requests, Dashboard
Ingredient Mixer	Raw material preparation	Mixer (Requisitions), Warehouse View, Dashboard
Production Sup.	Line management and oversight	WO Manage (partial), Incident Investigate, QC View, Maint. View, Dashboard
Factory Manager	Overall operations manager	All operational modules + Admin Users/Reports
Planner	Production scheduling	WO Manage, BOMs, Dashboard
Administrator	System administration	Full system access including all admin modules

The full permission matrix (Figure 2) shows the complete mapping of roles to submodules. Legend: R = Read/View, W = Write/Manage, X = Execute, A = Full Admin, - = No Access.

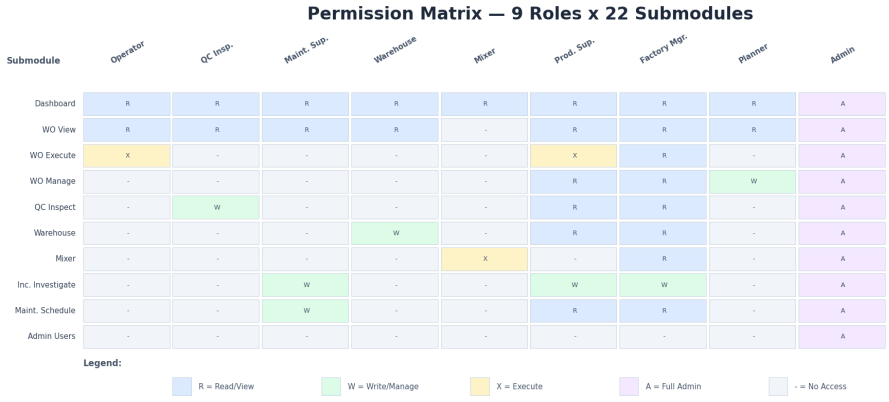


Figure 2: Permission Matrix — 9 Roles x Key Submodules

5. Workflow Diagrams

5.1 Work Order Lifecycle

The Work Order lifecycle defines the stages a production job passes through from creation to completion. Each stage has specific entry criteria, actions, and exit conditions. Quality checks and maintenance clearance are mandatory gates before production can proceed.

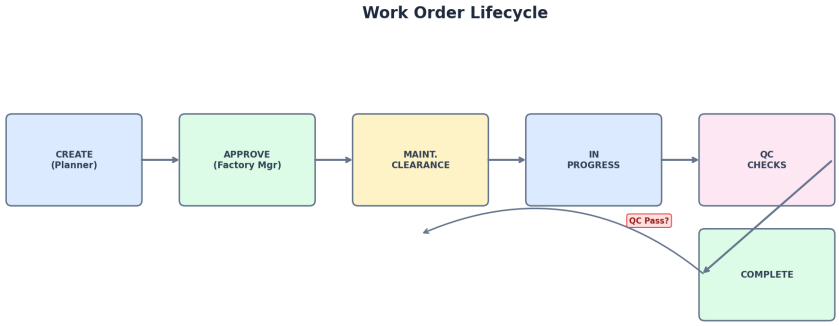


Figure 3: Work Order Lifecycle — From Create to Complete

Lifecycle Stage Details:

Stage	Entry Criteria	Actions	Exit Criteria
Create	Planner defines product, qty, line, schedule	Validate BOM exists, check line availability	Submitted for approval
Approve	WO in Pending status	Factory Manager reviews and approves	Approved or Rejected
Maint. Clearance	WO is Approved	Maintenance Supervisor runs pre-start checklist	All checks Pass
In Progress	WO is cleared for production	Operator executes production stages	Target quantity reached
QC Checks	Production complete	QC Inspector runs parameter checks	All checks Pass
Complete	QC checks passed	System updates inventory, records batch data	WO archived

5.2 BOM Requisition Flow

The BOM Requisition workflow connects the Planner's BOM approval to the Ingredient Mixer's material preparation and delivery. Each step is tracked with timestamps and user attribution.

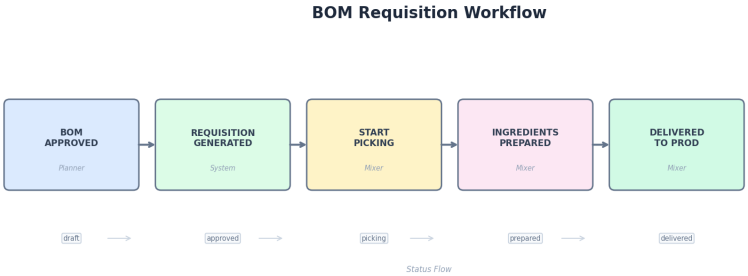


Figure 4: BOM Requisition — From Approval to Production Delivery

Step-by-Step Detail:

Step	Action	Actor	System Record
1	Planner approves BOM for a WO	Planner	Requisition created with status 'draft'
2	System generates pick list	System	Pick list with ingredient details auto-populated
3	Mixer starts picking	Ingredient Mixer	Status changes to 'picking', timestamp recorded
4	Mixer fetches and verifies ingredients	Ingredient Mixer	Lot numbers and expiry dates verified
5	Mixer marks prepared	Ingredient Mixer	Status changes to 'prepared'
6	Mixer delivers to production	Ingredient Mixer	Status changes to 'delivered', handover recorded

5.3 Incident Investigation Flow

When an incident is reported, it goes through a structured investigation process. Only users with 'Investigate' permission can access the full investigation workflow.

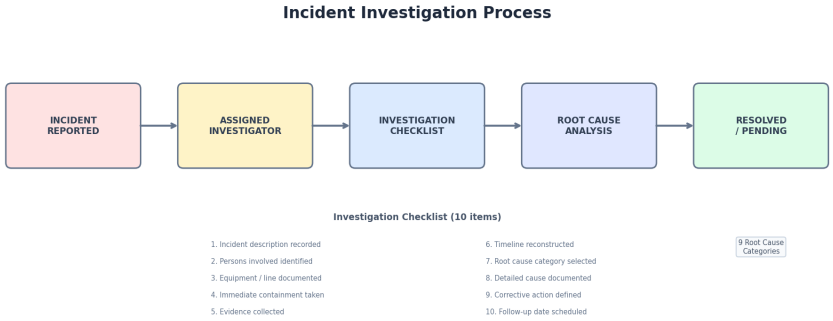


Figure 5: Incident Investigation Process

Investigation Checklist (10 items):

1. Incident description recorded
2. Persons involved identified
3. Equipment / line documented
4. Immediate containment taken
5. Evidence collected
6. Timeline reconstructed
7. Root cause category selected (from 9 predefined options)
8. Detailed cause documented
9. Corrective action defined
10. Follow-up date scheduled

6. Data Model Summary

The MES data model defines the core entities and their relationships. All interfaces are implemented in TypeScript for type safety and are ready for backend API integration.

6.1 Core Entities

Entity	Key Fields	Relationships
User	id, name, email, role, permissions[], active	Role, WO (assigned)
Role	id, name, description, permissions[], builtIn	User (many)
WorkOrder	id, productId, variantId, qty, line, status, schedule	Product, QCCheck[], Incident[]
Product	id, name, sku, category, variants[]	WorkOrder, BOM
BOM	id, productId, variantId, items[]	Product, Requisition
BOMRequisition	id, woId, status, items[], preparedBy	WorkOrder, BOM
QCCheck	id, woId, parameter, value, result, checkedBy	WorkOrder
QCForm	id, name, fields[], category	QCCheck
Incident	id, line, type, severity, status, assignedTo	WorkOrder, Investigation
Investigation	id, incidentId, checklist[], rootCause, action	Incident
MaintCheck	id, woId, type, items[], result	WorkOrder
MaintTask	id, equipmentId, type, schedule, status	Equipment

6.2 Status Enumerations

Entity	Status Values
WorkOrder	Pending, Approved, In Progress, Completed, Cancelled
BOMRequisition	Draft, Approved, Picking, Prepared, Delivered
QCCheck	Pending, Pass, Fail
Incident	Open, Investigating, Resolved, Closed
MaintTask	Pending, In Progress, Completed, Overdue

6.3 File Structure

Frontend (React/Vite):
• src/pages/ — Page components (Dashboard, WorkOrders, QC, Incidents, etc.)
• src/components/ — Reusable UI components (Layout, Cards, Tables)
• src/stores/ — Zustand state stores (appStore, authStore)
• src/data/ — Mock data and TypeScript interfaces
• src/lib/ — Utility functions and helpers

Configuration:
• vite.config.ts — Build configuration with path aliases
• tailwind.config.ts — Theme, colors, dark mode
• tsconfig.json — TypeScript compiler options